Engineer HAVN cabins for measured rest, not gadgets

The health value of a HAVN cabin sits almost entirely in four measurable conditions — **darkness below 1 lux, noise at or below 30 dB(A) at the pillow, CO2 held under about 800 ppm by the Panasonic ventilation unit, and air at 18–24 °C with a cool bed** — plus a disciplined light sequence (warm low light in the evening, true dark at night, a 30-minute dawn ramp before waking). These are the only interventions with strong or moderate objective evidence, they are cheap, and they are exactly what the best sleep operators (Zedwell, Equinox) sell. A virtual window is a worthwhile mood and anti-claustrophobia feature, but only three technologies fit a 100 mm wall: Innerscene's 84 mm Circadian Sky, a 46 mm edge-lit light box that a Guangzhou OEM can build for roughly $100–600, or a thin LCD "window"; the celebrated CoeLux-type sun-and-sky skylights need 190–270 mm of depth and cost $6,000–17,400, so they are a ceiling-mounted premium option at most. For the Health Pod the honest finding is that the best-evidenced 20-minute products are a short nap with a controlled wake and guided breathing audio; the red-light panel has moderate (low-quality) evidence for muscle recovery and mood, weak evidence for sleep, and no evidence for "longevity", and any skin or medical claim would pull it into EU MDR Annex XVI as a class IIb device. For Soundbox the binding constraints are the EU formaldehyde limit of 0.062 mg/m³ in force since 6 August 2026, EN 13501-1 B-s1,d0 fire classes for every lining, a CE technical file under the Low Voltage, EMC and RoHS rules (with HAVN becoming the EU "manufacturer" the moment it brands the cabins), and German building law that treats a 2.2 m-high box as furniture inside an approved room rather than as a guest room — which, together with Berlin's 12-guest-bed threshold for hotel-type regulation, also limits how many Double cabins a station should hold.

How to read the evidence grades in this report. **Strong** means multiple randomised trials or a consistent meta-analysis; **moderate** means at least one good trial or meta-analysis but small, heterogeneous or outside the cabin use-case; **weak** means pilot studies, physiological surrogates or case series; **marketing** means no peer-reviewed evidence was found. The research team could read most primary papers only through search extracts because direct page fetches were blocked, so figures taken from press or vendor summaries rather than the paper itself are flagged in the text; where a number could not be sourced it is omitted rather than guessed.

Darkness, quiet, fresh air and a cool bed carry most of the evidence

Ranked by strength of objective evidence and by what a self-service cabin can actually deliver, the sleep-environment levers fall in this order: envelope acoustics and blackout are foundational; CO2-controlled ventilation produces small but repeatedly replicated gains; room temperature in the right band is worth 5–10 % of sleep efficiency; bed-level cooling has a plausible mechanism but no trial; weighted blankets and lavender have moderate subjective effects in specific populations; sound masking has the weakest evidence but no reported harm. Everything else is comfort.

Quiet: build to hotel standard, not office-pod standard

The WHO community-noise guidance recommends **less than 30 dB(A) in bedrooms at night** (WHO/Europe noise fact sheet), and the 2018 European guidelines add source-specific night-time façade limits of 40–45 dB Lnight (RIVM summary of WHO values). German hotel practice is the right construction benchmark: DIN 4109 requires **R'w 47 dB for walls between guest rooms with a 32 dB corridor door, and its "enhanced" Beiblatt 2 level asks for R'w 52 dB walls with 37 dB doors**, with the explicit warning that the door drags the combined value down and must be checked together with the wall (bau.de

forum on DIN 4109 hotel rooms; Garant door handbook). Office pods are rated differently and perform worse: Framery, the market reference, reports speech-level reduction of just over **30 dB under ISO 23351-1** (Framery soundproofing page; Framery test press release). The arithmetic matters for site selection: a 30 dB pod in a 60 dB(A) office lobby lands at about 30 dB(A) inside, which just meets the WHO value, but the same pod in a 70 dB(A) station concourse sits near 40 dB(A), ten decibels over target. A station deployment therefore needs roughly 10 dB more attenuation than a standard office pod plus masking, and the HAVN door and the two ventilation penetrations are the weak points to engineer first. The ventilation unit itself is the probable noise floor in a quiet cabin, and HVAC noise is a documented satisfaction killer: a 2021 hotel study found sleep satisfaction strongly predicts overall satisfaction and is undermined by linen, pillows and **noise from the air-conditioning unit or heater** (PMC10130565). Specify silencers on supply and exhaust, run the Panasonic unit at the lowest speed that holds CO2, and publish the measured dB(A) at the pillow.

Broadband masking is a cheap hedge rather than a therapy. A 34-study review (1,103 people) found positive sleep effects in only 6 of 18 white-noise studies but in 9 of 11 pink-noise studies, concluded there is no strong evidence for auditory stimulation, and reported no adverse effects from short-term use (Capezuti et al., PubMed 34964434); a Sleep Medicine Reviews analysis graded the whole literature as low quality (McGill OSS summary of Riedy et al.). The practical design is a single small speaker playing pink or brown noise, default off, user-selectable, because it also raises the floor against transient corridor noise. **Grade: acoustics strong; masking weak.**

Cool room, cooler bed

The best thermal dataset is ~11,000 person-nights from about 50 older adults with wearables and bedroom sensors: sleep efficiency was best at roughly **20–25 °C, fell 5–10 % as the room warmed from 25 °C to 30 °C, and also declined below 20 °C**, with large individual variation (WebMD on Baniassadi et al. 2023). Sleep clinicians advise cooler, 15.6–20 °C (Sleep Foundation), and a 2023 Journal of Thermal Biology study showed heavier bedding widens the comfortable room-temperature range while a thin duvet narrows it (Sleep Foundation news summary). The cabin should therefore offer a **user setpoint of 18–24 °C with a 20 °C default** and a heavier duvet option. Bed-level cooling conditions the microclimate rather than 11 m³ of air and is the one feature hotels cannot easily retrofit: Eight Sleep's pad cools to 13 °C per side (Eight Sleep DE), the Chilipad Dock Pro (now Chilipad 1.0) runs 13–46 °C with water (Chilipad on Wikipedia) and BedJet 3 blows 19–40 °C air (Sleepme comparison). None publishes wattage and no randomised trial of these pads was found, so **grade: room temperature moderate; bed cooling plausible but unproven**. Thermally, a 10.6–12.7 m³ cabin with a sleeping occupant, 20–50 W of electronics and 36 m³/h of ventilation needs on the order of 300–600 W of sensible cooling, rising towards 900 W if the ventilation unit runs at full flow without heat recovery; a Peltier system would need roughly twice that in electrical input, so a water-loop pad or a small split with a heat-rejection path into the host building is the efficient route (derived from the figures above; no cabin thermal study exists).

Fresh air: let CO2 drive the Panasonic unit

EN 16798-1 Category I calls for about **10 l/s (36 m³/h) of outdoor air per person**, which corresponds to roughly 550 ppm CO2 above outdoor levels at the standard's sedentary emission rate (REHVA Journal, Olesen); Eurovent calls under 900 ppm "recommended" (Eurovent IAQ flyer) and WELL v2 caps occupiable spaces at 900 ppm with points for holding 750 ppm (Kaiterra on WELL v2). The bedroom-specific evidence points lower. A DTU climate-chamber study of 36 young adults found that relative to 750 ppm, **sleep efficiency fell 1.3 % and 1.8 % and time awake rose 5.0 and 7.8 minutes at 1,000 and 1,300 ppm**, with less deep sleep at 1,300 ppm (DTU Orbit); an earlier dormitory study with a fan triggered at 900

ppm (835 vs 2,395 ppm average) improved actigraphic sleep efficiency and next-day performance (DTU Roomvent 2014 paper); and a five-night study found 717 vs 1,150 ppm associated with better sleep depth and fewer awakenings (Mishra et al. 2018 record). DTU's review concludes bedroom CO2 should stay **below 1,000 ppm and preferably below 800 ppm** (DTU Orbit review), as does an ASHRAE Journal column (ASHRAE Journal, Wargocki).

HAVN's specified >180 m³/h unit is generous: 36 m³/h is 3.4 air changes per hour in the 10.6 m³ Single Cabin and two occupants in the 12.7 m³ Double need 72 m³/h (5.7 ACH), whereas full flow is 14–17 ACH. The unit can therefore idle at roughly 20 % (quiet) and boost on demand. The recommended control logic is **boost at 750 ppm, alarm at 1,000 ppm**, with a full-flow purge during every reset, which clears 15 cabin volumes in about ten minutes. Add HEPA plus activated carbon on the supply to meet WELL's enhanced PM2.5 target of 10 µg/m³ (Kaiterra). Exclude ionisers, "plasma" and ozone devices: a 2021 Building and Environment test of an in-duct bipolar ioniser found almost no air-quality benefit and new by-products including acetone and formaldehyde pathways (Illinois Tech on Zeng et al. 2021), and Germany's indoor-air commission at UBA lists ozone, plasma and ionisation among methods it does not endorse while noting mobile purifiers cannot replace ventilation (UBA IRK statement 2020). A 40–60 % relative-humidity band is conventional practice but no specific EN 16798-1 clause was retrieved, so treat it as a design target rather than a cited requirement. **Grade: moderate, with the only objective polysomnographic replication in this whole list.**

Dark: below 1 lux, with no glowing status LEDs

One night of moderate ambient light during sleep (about 100 lux versus under 3 lux, per a secondary summary) raised heart rate during sleep and worsened next-morning insulin resistance in healthy adults (Northwestern News on PNAS 2022), and the 2022 expert consensus sets the sleep environment at **≤1 lux melanopic EDI** (CET summary of Brown et al. 2022). A windowless cabin gets this for free; the risk is self-inflicted light from indicator LEDs, a control tablet, a glowing edge-lit "window" or an LCD backlight floor. The specification is **<1 lux at the pillow, all status LEDs off or deep red, and a hard-off for the virtual window during the sleep window**. **Grade: strong consensus, single-night physiological trial.**

Bedding, scent and weight: opt-in extras with a hygiene cost

A chain-weighted blanket produced a ~60 % response rate versus 5.4 % for a light control blanket over four weeks in 120 adults, but all had insomnia plus a psychiatric diagnosis (Natural Medicine Journal on E kholm et al. 2020; response figures via Onmanorama), and a crossover trial in autistic children found no objective improvement (Acibadem summary); a 6–8 kg blanket also has to be washed between guests, so offer it as a booked extra, not standard kit. Lavender inhalation improved sleep quality with a pooled **SMD of −0.56 across 11 RCTs (628 adults)**, with the authors cautioning about trial quality (PMC12904233); the acute single-session effect is unknown and allergy consent is needed, so a cartridge diffuser the guest switches on, with the reset purge removing residue, is the right implementation. The perception literature gives three cheap wins against claustrophobia in a 4.8–5.8 m² room: lighter ceilings and upper walls make rooms look taller (Oberfeld, Hecht & Gamer 2010), cornice uplight on the upper wall increases perceived ceiling height (CMS conference paper), and clutter makes rooms feel cramped (NSF-hosted spaciousness paper): matte light ceiling, darker floor, cove uplight, recessed controls and a low bed. **Grade: weighted blanket moderate (clinical population only); lavender moderate (low-quality trials); spaciousness levers weak-to-moderate.**

Parameter	HAVN target	Basis	Grade
Noise at pillow	LAeq,night ≤30 dB(A); envelope to DIN 4109 Beiblatt 2 (wall R'w ≥52 dB, door Rw ≥37 dB); duct silencers; publish measured dB(A)	WHO; DIN 4109	Strong
Room temperature	User setpoint 18–24 °C, default 20 °C; heavier duvet option; bed-level cooling pad optional	Baniassadi 2023; Sleep Foundation	Moderate
CO2	ERV boost at 750 ppm, alarm 1,000 ppm; ≥36 m³/h per person; reset purge at full flow	EN 16798-1 Cat I; DTU; ASHRAE	Moderate
Particulates	PM2.5 ≤10 µg/m³ on supply (HEPA + carbon); no ioniser/ozone	WELL v2 A05; UBA IRK	Standards-based
Light during sleep	<1 lux at pillow; ≤1 lx mEDI; status LEDs off or deep red	Brown 2022; PNAS 2022	Strong consensus
Humidity	40–60 % RH (conventional; clause not retrieved)	—	Unverified

A four-channel light recipe does the circadian work

The only consensus numbers that exist are from the 2022 expert recommendations: **≥250 lux melanopic EDI at the eye during the day, ≤10 lx mEDI from about three hours before bed, and ≤1 lx mEDI in the sleep environment** (10 lx only if night-time vision tasks are unavoidable) (PubMed 35298459; CET summary). WELL v2's circadian feature uses the same metric, commonly cited as 150 EML (~136 mEDI) for one point and 275 EML (~250 mEDI) for three, measured vertically at eye level (Innerscene WELL page), though one study guide gives 240 EML for the upper tier (WELL AP guide). Converting mEDI to the photopic lux a supplier quotes requires the melanopic daylight efficacy ratio of the actual LED, roughly **0.48 at 2700 K, 0.71 at 4000 K and 0.95 at 6500 K** for high-CRI sources (Zumtobel LFE); because measured 4000 K LEDs span 0.42–0.72, the HAVN specification should require a measured spectral power distribution and computed mEDI per channel, not a CCT alone (CLTC/IES LD+A 2019).

Scene	mEDI at eye	CCT	Photopic lux at eye	Note
Daytime arrival / reading / nap exit "energise"	≥250	4000–6500 K, CRI ≥90	~265–350 lx	Brown daytime floor; 5–15 min at nap exit
Evening arrival (<3 h before sleep)	≤10	2200–2700 K	≤~20 lx ambient; warm localised reading light	Keep reading light close to page
Wind-down (last 20–30 min)	1–5	amber, <2200 K	a few lux	No saturated red at intensity
Sleep	≤1	—	<1 lx	Hard-off for window and tablet
Mid-night navigation	≤10	deep red / amber at floor level	5–20 lx of long-wavelength light	Floor cue lines, not overhead
Dawn wake (overnight stays)	ramp to 100–250	2000 K → 4000–5000 K	0 → 100–300 lx over 30 min	Compress to 5–10 min for naps (untested)

Two engineering consequences follow. A tunable-white strip whose warm end is 2200–2700 K cannot get below about 0.45 MDER, so delivering night-navigation light under 1 lx mEDI at usable brightness needs a **separate amber/red channel** — a four-channel (tunable white plus amber/red) 24 V system, not a two-channel one. And the research dawn protocols converge on a **linear 30-minute ramp from under 1 lux**

to 100–300 lux (PMC270037; Gabel et al. 2013 PDF citing van de Werken 2010), with van de Werken finding the gradual ramp beat acute light at the alarm, so the cabin must ramp before the alarm rather than switch on bright light at it.

The dawn-simulation evidence is consistent but small: a home device significantly reduced sleep-inertia complaints (Innerscene summary of Giménez et al.); 17 sleep-restricted adults showed better mood, well-being and cognition after dawn simulation than after dim or blue light (Gabel et al. 2013); an eight-person study found faster post-waking skin-temperature decline but no cortisol or performance effect (Thompson et al. 2014); and a 2024 review calls the countermeasure literature "mixed" for methodological reasons (PMC10969141). The best controlled test of a multimodal "smart alarm" (light, sound, temperature) found **little overall objective impact**, with benefits depending on chronotype and on light exposure lasting more than 15 minutes (Campanella et al. 2024). Market the dawn ramp as "wake more gently", never as "proven performance". **Grade: moderate, subjective outcomes.**

Red light at night deserves a specific warning because it is widely mis-sold. Saturated red through closed eyelids mitigated sleep inertia on waking in a 30-person crossover (PubMed 31118850), but its lead author says she "would not make the claim that red light promotes sleep" (KSL interview with Figueiro), and a 2023 review reports red light **increases subjective alertness, anxiety and negative emotions in both healthy people and insomnia patients** (Frontiers in Psychiatry 2023). Use dim amber/red for navigation and for the last minute before a nap wake, not as a pre-sleep "red light therapy" scene. For fall-safe night movement, horizontal and vertical light-cue lines at floor level improved postural control in older adults versus plug-in night lights (Figueiro et al. 2012).

Compliance for the lighting is straightforward but must be specified: EU Ecodesign Regulation 2019/2020 limits flicker to **PstLM ≤1.0 and stroboscopic visibility to SVM ≤0.4** for mains light sources (Fagerhult); luminaires with separate drivers escape the letter of the rule, so HAVN should impose the same limits contractually, with constant-current reduction or >3 kHz PWM at the deep dim levels used at night. Photobiological safety follows EN 62471, where risk groups RG0 and RG1 need no measures and RG2 or above must be marked on the product and manual (Delta Light photobiological safety; ERCO). On components, Shenzhen Sunricher makes both Casambi RGBCCT controllers (Casambi ecosystem) and the DALI-2 DT8 tunable-white drivers that European consultants expect (its SRPN-9105 part appears in Simpex's catalogue, Simpex DT8 driver); LEDGEAR is a second mainland-China DALI-2 DT8 source (HK TDC listing); and sunlight-spectrum LEDs such as Seoul's SunLike (Business Wire via ANSA) cost a few dollars per COB at volume (Dasenic listing). The whole four-channel BOM for a cabin is in the low hundreds of dollars; the controller should run scenes locally with no cloud dependency and expose DALI or a Casambi API for building integration.

Only three window technologies fit a 100 mm wall

A "virtual window" means one of four things. Rayleigh-scattering skylights place a polymer sheet loaded with titanium-dioxide nanoparticles in front of a white LED so blue light scatters (sky) while the direct beam stays warm (sun) (Core77); the sun illusion requires distance between source and scattering panel, which is why these units are deep (US patent 12000583). The optical sheet was for years exclusive to Italy's CoeLux but Chinese trade press now lists domestic makers including Changzhou FS Optronics, Opple, Yeelight, Pak and Shenzhen Yichuang (OFweek Lighting). The second family is thin tunable-white "sky" panels without a sun; the third is display-based windows; the fourth is backlit photographic light boxes. Depth is the decisive filter for HAVN's 100 mm panel:

Product	Technology	Depth	Power	Range / control	Price indication	Fits 100 mm wall?
Innerscene Circadian Sky	Tunable sky, no sun	84 mm plenum	35–75 W	2,200–40,000 K, CRI 91+; 300×600 to 600×1200 mm	Not public (quote only)	Yes, flush (Innerscene FAQ; I+S Design)
Sky Factory Luminous Virtual Window	Edge-lit photographic light box	46 mm	low tens of W	Static image; tunable only with TW driver	~$105/sq ft (2008 figure)	Yes, surface or flush (LVW guide; SF Chronicle)
Atmoph Window 2	27" FHD LCD, 15-min loops	~30–60 mm	n/p	Subscription content	$380–448	Yes (Atmoph store)
Chinese OEM "blue sky" 1200×600	Rayleigh / edge-lit clones	Not published	72–120 W	App dimming	$60–600 (RYOPT $535 at 2–49 pcs, $470 at 1,000+)	Unknown — verify (Alibaba RYOPT; Alibaba 30×120 cm)
Mitsubishi Misola	Rayleigh sky, asymmetric lit frame fakes sun angle	~120 mm	n/p	Dawn/sunset/night scenes, schedule	$6,000–7,000 (2020)	20 mm bump-out; wall version exists (Techbang; Boing Boing)
Yeelight Pro P21	Consumer Rayleigh sun+sky	190 mm (≥200 mm void)	80 W	1,600–8,000 K, 2,100 lm	€373	Ceiling only (homescreen.pl)
Innerscene A7 / Virtual Sun	Rayleigh sun+sky	254–267 mm	8–200 W, 5,600 lm	Sun 2,500–20,000 K	~5–6 units per CoeLux LC45 price	Ceiling only (Innerscene comparison; LessWrong thread)
CoeLux HT25	Rayleigh sun+sky	250 mm	~200 W, 5,800 lm	DALI	$15,600–17,400	Ceiling only (Kazzar; 360lamp.ca; Baunetz Wissen)
LiquidView	Three 75" 4K displays	150–200 mm	high	24-h real-time views, $59–199/month	$2,999 player, $10,499–49,000 window	No (LiquidView; Leon Speakers)

Three integration facts for Soundbox follow from the table. First, a 450×1200 or 600×1200 mm panel mounted portrait fits inside one 80 cm-wide wall module, whereas a landscape 1200×600 spans two modules and puts a panel joint behind the luminaire. Second, cutting an 84 mm luminaire into a 100 mm acoustic sandwich removes most of the mass and absorber over that area and creates an acoustic weak spot in a wall that must reach hotel-grade attenuation (our inference; no vendor publishes acoustic data for recessed panels); a 46 mm surface-mounted light box, or placing the window in the ceiling panel where the "outside" is the host room rather than a neighbouring cabin, avoids this. Third, a true sun-and-sky unit is possible only in the ceiling with its housing protruding 200–270 mm above the cabin roof, which a room-in-room installation can accommodate if the host building has more than about 2.5 m clear height; the old CoeLux LC45 recommended 2.44–2.6 m ceilings for the geometry to work (CE Pro). A 200 W unit also dumps the heat of roughly two adults into a sealed 10.6 m³ box, whereas the 35–75 W Circadian Sky or a 20–40 W light box is easy for the ventilation to carry.

The evidence for what a virtual window does is affective, not cognitive. In the key randomised lab study, 100 healthy adults in a windowless office under a Rayleigh skylight reported **lower tension, anxiety and claustrophobic symptoms and higher positive mood** than under fluorescent light — but also made riskier and more selfish decisions (Canazei et al. 2017). Ten men exposed to a Rayleigh skylight by day secreted more melatonin that night (Yasukouchi et al. 2019), which supports using the window as a daytime light dose rather than a night feature. A 2025 randomised crossover (n=32) found a tunable artificial window reduced visual fatigue and raised positive affect by 73 % during a 15-minute rest (Journal of Building Engineering 2025); a 2024 study found artificial windows matched real ones on mood and alertness but not on physiology, and warned that **over-lighting at eye level can hinder arousal, cognition and mood restoration** (Building and Environment 2024); its 2025 follow-up found dynamic, natural, well-rendered content performed best, with mood but not physiological restoration (Building and Environment 2025). A field experiment with nature-video windows raised mood and happiness without measurable productivity change (Wohn et al. 2020). The one large clinical dataset — 181 inpatients, 53 % less acute stress under a Sky Factory ceiling — is vendor-reported and unreplicated (Sky Factory Japan e News), and a registered ICU trial of a digital window never reported results (NCT05486780). **No study tests a virtual window in a sleeping room**, so any HAVN sleep claim would be extrapolation and is a good candidate for an in-house pilot measurement. **Grade: mood and anxiety moderate; spaciousness, cognition and sleep weak or absent.**

The failure modes are known. A Rayleigh unit's "night sky" scene is still short-wavelength scattered light and must be off during sleep. Edge-lit boxes and LCDs have a brightness floor, so they need a hard-off relay rather than dimming. Head-tracked parallax works for one viewer only and scenes without parallax feel "less believable" (Journal of Perceptual Imaging); loops become visible, which is why Sky Factory runs unedited 30-minute-to-hours sequences and LiquidView films 24-hour views (Sky Factory Japan 2010; TechNewsWorld). A ceiling or high-wall placement is the least uncanny and avoids the eye-level over-lighting problem. For EU sale, an OEM panel needs CE under the Low Voltage and EMC directives, RoHS, an EN 62471 report and Ecodesign flicker data; none of the Alibaba listings mention EN 62471 or SVM data, and whether Chinese Rayleigh sheets infringe CoeLux patents in the EU was not researched and should be asked. The practical strategy is a mixed pilot — one or two Circadian Sky 450×1200 units as a quality reference plus two to five Chinese panels ($100–600, MOQ 2–5) to benchmark illusion and compliance — and, for 2,500 cabins, an OEM edge-lit tunable light box or sky panel at an implied $100–500 per unit, with the compliance test package negotiated into the contract.

The Health Pod's best 20 minutes are a nap and a breath, not a machine

The nap literature sets the protocol. In the classic duration study a 10-minute nap gave the most immediate and sustained benefit, a 20-minute nap's benefits appeared about 35 minutes after waking and lasted up to 125 minutes, and a 30-minute nap caused sleep inertia (secondary summary of Brooks & Lack 2006). NASA's cockpit study found crews fell asleep in about 5.6 minutes and slept 25.8 minutes with improved alertness and performance (NASA NTRS 19950006379); an 18-study shift-work meta-analysis (494 participants) found a performance effect size of **0.25 (95 % CI 0.10–0.41), mainly ≥30 minutes after waking**, with memory worse immediately on waking (HAL record of Dutheil et al. 2020). A structured review found no convincing reactive countermeasure to sleep inertia, with caffeine "perhaps the best option" when taken before sleep (Hilditch et al. 2016), and a 2022 NASA update suggests blue-enriched light on waking helps mainly after night-time slow-wave sleep (NASA NTRS 20220012907). The resulting **20-minute Power Nap protocol**: two minutes of settling with a 60-second slow-breathing cue, lights to near-dark, optional lavender, lying time capped at 20 minutes so actual sleep is 10–15 minutes, wake by rising audio plus brightening light over 60–90 seconds, then a two-to-three-minute upright "re-boot" under

≥250 mEDI daytime light before exit, with caffeine offered at the kiosk *before* the session for an "alert" variant and no promise of alertness in the first 30 minutes after waking. A **90-minute Full Cycle** variant for the sleeping cabins uses a ±10-minute movement-based wake window instead of a fixed alarm. **Grade: strong.**

Intervention	Fits 20 min, unattended?	Grade	Key evidence
Short nap 10–20 min with controlled wake	Yes	Strong	ES 0.25 (Dutheil 2020); NASA 25.8 min nap (NTRS)
Caffeine before nap	Yes (kiosk)	Moderate	Caffeine + nap reduce drowsy driving (SWOV); caffeine best before sleep (Hilditch 2016)
Resonance breathing / HRV biofeedback audio	Yes	Strong as training; moderate acute	g = 0.81–0.83 for stress/anxiety, 24 studies (Goessl et al. 2017)
Yoga nidra / NSDR audio, 11 min	Yes	Moderate (acute stress); weak (sleep)	362-person online RCT (Moszeik et al. 2025)
Guided mindfulness audio	Yes	Moderate, industry COI	g = 0.46 vs controls (protocol citing meta-analysis)
Dawn / blue-enriched wake light	Yes	Moderate, subjective	See lighting section
Red/NIR panel — muscle recovery, endurance	Yes	Moderate (very-low-to-moderate quality)	39 RCTs, 861 participants (Vanin et al.)
Red/NIR panel — mood, drowsiness, resting HR	Yes	Weak-to-moderate	Only the 6.5 J/cm² dose helped; no sleep effect (Giménez et al., PubMed 36671752)
Red/NIR panel — sleep, melatonin	—	Weak	One 20-athlete non-randomised cohort (PMC3499892); double-blind RCT null on sleep
Red/NIR panel — skin, anti-ageing	—	Weak, plus MDR class IIb risk	Pre-2018 split-face trials only (PubMed 17566756)
Red/NIR panel — longevity, mitochondria, inflammation	—	Marketing	Only a 20-person morning 670 nm vision study (EurekAlert)
Binaural beats in audio	Yes	Weak-to-moderate	g = 0.45 across 22 studies, mixed later (PubMed 30073406)
Vibroacoustic transducers in bed	Yes	Weak	HRV surrogate in 54 students (Chiba symposium paper); mostly case reports (Music & Medicine)
Warm (FIR) mat during pre-nap minutes	Partly	Moderate by extrapolation	Warm bath 40–42.5 °C 1–2 h before bed shortens latency (Haghayegh et al. 2019)
PEMF mat	Opt-in only	Weak; hard contraindication	Pacemaker contraindication on CE label (Orthofix manual); whole-body evidence "insufficient" (Superpower PEMF guide)
Compression boots	No (30–60 min sessions)	Weak	17 studies, 319 people, "lacks strong support" (Maia et al. 2024)
Closed-loop acoustic slow-wave enhancement	No (needs EEG and N3)	Moderate and declining	g = 0.25 (Wunderlin 2021); effect fell to d = −0.39 (Harlow 2023)

Intervention	Fits 20 min, unattended?	Grade	Key evidence
Sensate, Somadome/OpenSeed pods as such	—	Marketing	No validation (WellnessPulse)
Cold plunge, oxygen bar, grounding mat	No	Weak / marketing, safety issues	Cold-shock risk needs supervision (Portsmouth guidance); oxygen is a prescription drug in the US (Oxygen bar)

HAVN's 630–850 nm panel matches the industry's 660/850 nm pattern, and the engineering questions are dose, heat and eye safety. Consumer and pro panels advertise roughly 85–160 mW/cm² at 15 cm, measured at the panel centre: a third-party test put the PlatinumLED BioMax 900 at 87 mW/cm² at 6 inches (Dr. Axe panel ranking), and that half-body panel draws **490 W and weighs about 15 kg** (frontdeskreview tracker). At a true 50 mW/cm², 20 minutes delivers 60 J/cm², whereas the only dose with consistent well-being benefit in a double-blind trial was 6.5 J/cm², reached in one to two minutes; a 10-minute light phase followed by 10 minutes of rest or breathing is therefore a better-founded 20-minute design than 20 minutes of full irradiation. A 450–490 W panel running 15 minutes dumps about 400–440 kJ into a 10.6 m³ room, enough for a ~30 K air-temperature rise if nothing were removed, so the ventilation must boost to roughly 50–100 m³/h during and after a session and the panel's fans will add noise. Mount a half-body panel (about 91×30 cm) on the long wall beside the therapy bed at a fixed 20–30 cm, require a nine-point irradiance map at that distance from a calibrated spectroradiometer rather than a centre reading, and obtain an EN 62471 report that includes the separate retinal-thermal limit for 780–1400 nm sources, where the eye's aversion response does not protect (Osram/ams eye safety of IREDs); a lighting product tested RG2 at 20 cm and RG1 at 30 cm illustrates how distance moves the class (Hubbell case study). Timing runs against vendor advice: the UCL 670 nm vision effect appeared only with morning exposure (EurekAlert) and red light raises alertness, so sessions belong in the day or after exercise, not before sleep.

Branded panels retail at about **$419 (Kala Pro, half-size) to $1,299–1,699 (BioMax 900, MitoPRO 1500X, Joovv Solo 3.0)** (Sam's Club Kala Pro; frontdeskreview). Shenzhen OEMs list a 300 W 660/850 nm panel at **$278–305 FOB with MOQ 1 and 5–7 day lead time** (Beautetrade Idealight 300 W) and a 1500-class full-body panel at $1,618–1,698 (Beautetrade TL1500), with Dongguan RedThera and Shenzhen Ilux offering OEM/ODM; wattage labels on these listings are inconsistent ("1000 W" products with 1,800 W actual draw), so buyers must demand chip count, drive current and measured irradiance. Other Health Pod hardware is cheap relative to the cabin: a HigherDOSE PEMF Go Mat is $700–899 and 120 W (Northern Fitness), an infrared sauna blanket draws 450–500 W measured and tops out at 70 °C (Fit&Well), Normatec 3 boots cost $799–1,149 (T3), the OPUS SoundBed listed at $1,999 (Yanko Design); the expensive items are branded pods — Somadome at $14,500–35,000 (IndustryM; Leisure Opportunities), OpenSeed at $16,000 (Manufacturing Business Technology) and the MetroNaps EnergyPod at $8,000–13,000 (Unanswered guide). Everything fits on one 230 V, 16 A circuit.

"Longevity" can be framed honestly only through sleep behaviour, not hardware. In 60,977 UK Biobank participants, the four more-regular sleep-regularity quintiles had **20–48 % lower all-cause mortality** than the least regular, and regularity predicted mortality better than duration (PMC10782489); mortality follows a U-shape with its nadir near seven hours (Yin et al. 2017). Napping is a warning rather than a selling point: across 21 cohorts (371,306 people) nappers had a cardiovascular hazard ratio of 1.18, with **naps under one hour showing no excess (HR 1.03) but naps of an hour or more HR 1.37** (PMC11482734), and an earlier dose-response analysis found RR 1.82 for naps ≥60 minutes (PMC4667384). The defensible statements are "short naps under an hour are not linked to the cardiovascular risk seen with

long naps", "a consistent sleep schedule is associated with lower mortality in large cohorts", and "most adults do best near seven hours"; the product feature that follows is scheduling and a regularity score, with a hard cap of 30 minutes lying time in the nap product. Any "adds years", "reverses ageing" or disease-prevention wording is both unsupported and an MDR intended-purpose trap. For unattended safety, the design principle is that audio, light, breathing guidance and a plain bed carry near-zero liability, while PEMF (pacemaker contraindication), heat and compression are opt-in modules unlocked only after an in-app screening for implanted devices, pregnancy, epilepsy, clotting disorders and heat intolerance, with controller-enforced limits (surface temperature capped around 40–45 °C, no strobe or flashing light); these thresholds are design suggestions, not sourced limits.

Price anchors show a 20-minute Health Pod session at €25–45 is in-market: Munich's RedUp sells 15-minute red-light sessions from €24 (Urban Sports Club listing), Berlin cryotherapy starts at €24 (Hands Health Cooling), Somadome sessions average $50 for 20 minutes (Leisure Opportunities), Restore sells a $75 first-visit pass bundling cryo, red light and an hour of compression (Referrizer Woodbury), Remedy Place charges $495 a month (Spa Business), and longevity clinics start at €5,900 for a Lanserhof week (Strøm Spa blog).

Radar sensing can deliver a credible sleep report, if consent comes first

Contactless sensing is good enough for a hospitality report but not for clinical staging. The Withings Sleep Analyzer, the only CE-marked, FDA-cleared under-mattress product with EU retail availability (about **€120–150**, tink.de), reached 87 % epoch-level sleep-wake accuracy against home polysomnography in 117 healthy adults but only 73 % sensitivity for wake (Stefanos et al. preprint) and 88 % sensitivity and 88.6 % specificity for moderate apnoea in 118 patients, in a study partly authored by Withings staff (Edouard et al. 2021). Google's bedside 60 GHz radar scored the same 87 % overall but detected only 55 % of wake epochs (Google Research); the Emfit strip lost data in 47 % of lab participants and overestimated sleep by three hours (Kholghi et al. 2022), so it is unsuitable as a primary sensor. Shenzhen 60 GHz breathing-and-heartbeat modules cost €45–60 at single quantity (OpenELAB MR60BHA2), the R60ABD1 class reports bed occupancy and sleep status through bedding (Tindie), and Infineon's BGT60UTR11AIP radar-in-package with a Sleepiz algorithm partnership gives a Guangzhou integrator a reference design (Infineon press release). A radar that doubles as presence, breathing and bed-occupancy sensor also handles reset detection without a camera. The defensible report content is time in bed, estimated sleep, time to settle, awakenings, resting and lowest heart rate, respiration rate, snoring minutes and loudest dB(A), and the cabin's own CO2, temperature and noise record — with deep/REM labelled "estimated" and any breathing irregularity phrased as a prompt to see a doctor. The Double Cabin's 160 cm bed is a limitation here, because under-mattress and single-beam radar sensors are single-sleeper devices.

Only Nine Hours has productised this: its capsules sense movement, snoring and facial images with consent, email a report, and refer guests with poor metrics to hospitals (Nine Hours sleep analysis), and its 70-capsule NTT Data Sleep Lab adds infrared cameras, microphones, heart rate and breathing (Travel Voice). In the EU that camera-based design would be hard to justify: the Article 29 Working Party treated raw sensor data from which health conclusions can be drawn as health data (Hunton on the WP29 annex), and the CJEU's October 2024 Lindenapotheke ruling confirmed a broad reading of "data concerning health" in which neither the accuracy of the inference nor the controller's intent matters (Bird & Bird). The design consequences are tracking off by default and switched on per stay by explicit consent, no cameras, microphone processing on-device for snoring minutes and dB only, pseudonymised stay IDs with short retention and EU hosting, a data-protection impact assessment before the Berlin pilot, and comparison only against the guest's own previous stays. The surrounding controller is commodity: a

Sensirion SCD41 CO2/temperature/humidity sensor at about $14–20 and an SPS30 particulate sensor at about $33 (Future Electronics), a Tuya Zigbee DIN-rail hotel room controller for lights, fan and heater (Best Buy CA listing), or Matter 1.3, which already has a certified air-quality device type (CSA product listing). The evidence-backed closed loops are three: pre-sleep dimming and cooling on a schedule (the Equinox pattern), CO2-driven ventilation, and a dawn or red-light wake ramp of at least 15 minutes; "wake in light sleep" should be a soft window, not a headline claim.

What Soundbox must document: emissions, fire, CE and claims

Formaldehyde and VOC: the legal floor moved in August 2026

REACH Annex XVII entry 77 bans placing furniture and wood-based articles on the EU market if formaldehyde released in the Appendix 14 chamber test exceeds **0.062 mg/m³, in force since 6 August 2026** (CIRS; TÜV SÜD); that is half the old E1 class in EN 13986 (Measurlabs). China's GB/T 39600-2021 grades panels **E1 ≤0.124, E0 ≤0.050 and ENF ≤0.025 mg/m³** (Intertek China), so a Chinese "E1" board fails the EU limit and "ENF" is not zero-emission (NBD); Chinese reports conflict on whether E0 has become mandatory under GB 18580-2025 (Beijing Daily 2026), which is why HAVN should ask for the test report, not the label. The credible specification for a "healthy, low-emission" cabin is ENF, or at minimum E0, boards with an ISO 17025 chamber report on the finished article, plus a full 28-day VOC report under the German AgBB scheme (2024 edition, UBA AgBB 2024) on the assembled wall, ceiling and floor build-ups, because German building law via MVV TB Annex 8 treats AgBB compliance as proof of health protection in occupied rooms (Baunetz Wissen; UBA Bundesgesundheitsblatt 2026). Any "antimicrobial" surface is a treated article under the Biocidal Products Regulation: only approved active substances may be used and a claim triggers mandatory labelling (BPR Chapter XIII), and silver-zinc-zeolite approval carries restrictions on skin-contact textiles (Implementing Regulation 2024/2635). Given that copper surfaces show only modest contamination reductions and no clear infection benefit in systematic reviews (Universidad de Navarra record), the antimicrobial claim is a marketing risk best dropped. Framery's build-up — recycled acoustic foam and felt sandwich with **B-s1,d0 foam and Cfl-s1 carpet** (HuntOffice Framery O) — is the market reference Soundbox should match or beat.

Fire, sprinklers and what a 2.2 m box is in German law

Germany has no product category "sleeping pod". Building materials are classed non-combustible, flame-retardant (schwerentflammbar) or normally combustible, and highly flammable materials are banned (MBO excerpt). Framery states its materials are tested to national or international standards and that the need for detectors or sprinklers inside a pod is decided by local authorities (Framery fire-safety help); its reseller cites an Inspecta statement that no sprinklers are needed inside its pods (HuntOffice Q&A). The realistic Berlin baseline is **EN 13501-1 classification reports from an EU-accredited lab (not Chinese GB 8624 reports) for every exposed lining, target B-s1,d0 with Cfl-s1 flooring**, EN 1021 ignitability for upholstery, and a written agreement with the host building's fire-safety engineer and insurer under the current sprinkler guideline VdS CEA 4001:2024 (VdS) on sprinkler pass-throughs or drop heads in the ceiling panel and a smoke detector wired to the building alarm. The door must open from inside without key or power at all times, with an electronic lock that releases on power loss and a main isolator reachable from outside.

Two thresholds shape the station design. Berlin's building code requires **2.50 m clear height for Aufenthaltsräume** per a Berlin parliamentary answer (Abgeordnetenhaus S18-24401) (a 2024 lobby submission quotes 2.40 m, Berlin lobby register PDF), so a 2.2 m cabin cannot itself be an approved room

and should be classified as furniture or equipment inside an already-approved room — the same logic as the Framery precedent — with written sign-off before installation. Berlin's Betriebs-Verordnung applies hotel-type rules to Beherbergungsstätten with **more than 12 guest beds** (GVBl. Berlin 2005), a threshold mirrored in Baden-Württemberg and NRW (LBO BW §38); eleven Single cabins are eleven beds, but each Double cabin counts two, so a station with more than one Double crosses 12 and invites escape-route, alarm and operational rules written for hotels. The host building is very likely already a Sonderbau in Berlin if any office room exceeds 400 m² or a storey exceeds 1,600 m² (BauO Bln §2). Where employees use the cabins, workplace rule ASR A4.2 requires rest couches to be upholstered with a washable or disposable cover and privacy preserved (KomNet 42495).

CE marking: who is the manufacturer

A cabin with mains lighting, fans and electronics is an electrical product under the Low Voltage Directive 2014/35/EU and EMC Directive 2014/30/EU, with RoHS and — for any Wi-Fi, Bluetooth or NFC lock module — the Radio Equipment Directive; a typical declaration of conformity maps these to EN 62368-1, EN 55032/55035 and EN IEC 61000-3-2/-3-3 (Unify EU DoC), with EN 60598 for integrated luminaires, EN 62471 for all light sources and IEC 60335-2-65 for any built-in UV-C air unit (IEC PAS 63313 summary). The freestanding shell is not a construction product, so the CPR is not the CE route for the cabin as a whole, though wood panels and floor coverings carry their own declarations. The decisive point for the HAVN–Soundbox relationship is the EU definition of "manufacturer": **any person who has a product designed or manufactured and markets it under their own name or trademark** (Westwood Products summary of GPSR). If the cabins carry the HAVN brand, HAVN holds the technical file, signs the declaration, affixes the CE mark and keeps records for ten years; if they carry Soundbox's brand, HAVN is importer and Soundbox needs an EU authorised representative. Either way Soundbox should supply CB-scheme reports (IEC 60335/62368/60598) from TÜV, SGS or Intertek that an EU lab can convert, an EMC test on the complete assembled cabin with all devices running, a bill of electrical components with their own declarations, RoHS statements, Schuko wiring and the external isolator. One commercial caution: if HAVN has not registered its brand in China, the supplier may legally use or register it there (Harris Sliwoski).

The 20-minute reset is a logistics design, not an automation product

No operator automates linen. Napcabs, the closest German precedent at Frankfurt and BER, notifies cleaning staff automatically after each use, who remake the bed with fresh linen before the cabin is released (International Airport Review; Fraport). The workable HAVN cycle is: door-lock and radar confirm the cabin is empty; the ventilation unit purges at full flow for the whole reset; a zipped waterproof mattress and pillow encasement is changed on a long cycle while a single-piece fitted-sheet-and-duvet-cover "cassette" is swapped in under five minutes; and an optional 254 nm UV-C surface pass runs only while locked, with a lamp-off interlock on the door. The 254 nm wavelength does not generate ozone, unlike 185 nm sources (Holtop); if a cleaner could ever be exposed, Directive 2006/25/EC's daily effective limit of 30 J/m² applies and the operator must assess exposure (sglux datasheet citing the Directive), which interlocks eliminate. Far-UVC at 222 nm is not yet "set and forget": a 2024 safety review supports use within guidelines but calls the upper limit unresolved (PubMed 37929787), and filtered KrCl lamps generate limited but non-zero ozone (Ma et al. 2023), a concern in a small, poorly mixed volume. Shared contact items in the Health Pod — headphones, boots, mats — need disposable liners and wipeable non-fabric surfaces, mirroring the "between every client" standard for foot spas (NSW Health fact sheet).

Claims: intended purpose decides MDR, and the Health Pod is the exposure

MDR scope follows the manufacturer's stated intended purpose, and Annex XVI captures listed non-medical products regardless of claims, including **"high intensity electromagnetic radiation" equipment "for skin resurfacing, tattoo or hair removal or other skin treatment"**, now subject to Common Specifications under Regulation 2022/2346 (applicable since June 2023, transition to June 2028) and classified IIb by Regulation 2022/2347, while "equipment using infrared optical radiation to warm the body" is excluded (NAMSA; BSI white paper; QbD Group). Whether a ~100 mW/cm² LED panel counts as "high-intensity" is undefined in any source found and is the single most important regulatory question to put to a notified body or regulatory counsel; red and blue light devices that make therapy claims must comply with MDR in a consultancy's reading (Obelis), and the newly adopted EN IEC 60601-2-57:2026 for non-laser light equipment for medical and aesthetic use shows where the standards are heading (CEN-CENELEC). The precedents point both ways: Withings deliberately took its Sleep Analyzer through CE medical certification to claim apnoea detection (Med-Tech News), while German wellness-equipment maker Wellsystem declares its products "general wellness and fitness devices" outside MDR Article 2(1) (Wellsystem). The claim ladder for HAVN is therefore: safe — "quiet, dark, private space to rest", "fresh-air ventilated", "low-emission materials tested to AgBB and REACH", "20-minute power nap", "wind-down lighting"; borderline but defensible — "helps you feel more rested", "designed to support relaxation and recovery", "red light setting for relaxation"; out of bounds — "treats insomnia, jet lag, burnout", "improves recovery from injury", "lowers cortisol", "anti-ageing", "collagen", "longevity" tied to a mechanism, or any diagnosis language from sensors. HAVN must not add medical claims in its own app to third-party devices, or it assigns them a medical intended purpose and becomes their manufacturer. German advertising law (Heilmittelwerbegesetz, UWG) was not researched in this round and is an open item for counsel.

Overnight stays trigger registration duties

If guests pay to sleep overnight, the operation is a Beherbergungsstätte for registration purposes: under §§29–30 Bundesmeldegesetz the operator keeps a special registration form, and foreign guests must sign it by hand on arrival (lexetius §29 BMG; §30 BMG); since 1 January 2025 German nationals no longer complete it (Garmisch-Partenkirchen notice; IHK guidance). An unattended cabin satisfies the handwritten-signature rule only with a signature pad accepted by the Berlin Meldebehörde or a paper form; whether daytime-only naps count as accommodation at all, and whether city tax applies, were not resolved and should be checked with the Bezirksamt.

Operators sell darkness, quiet and clean air at €12–45 an hour

The feature set of comparable operators clusters into three tiers, and none of the public descriptions mention red light, virtual windows or scent.

Operator	Format	In-room kit	Price	Published sleep data
Napcabs (Munich, Frankfurt, BER)	Airport cabin, 200×80 cm bed, unstaffed	Touchscreen for light, temperature, alarm; air-conditioning; desk; human cleaning after each use	€17/h day, €12/h night, 2 h minimum (€34); XL €24.50/18 (Munich Airport)	None
Sleepbox (Washington Dulles)	Airport micro-room	Private room	$30–44/h (The Points Guy)	None

Operator	Format	In-room kit	Price	Published sleep data
Minute Suites (US airports)	Daybed suite	TV, shower at some sites	$48 first hour + $12 per 15 min; $140 overnight (FlightQueue; WFAE)	None
Zedwell capsules (London, 2025)	Capsule hotel	Individual ventilation and climate control, noise reduction, ambient light, Hypnos mattress	From £30/night (Globetrender)	None
Zedwell cocoons (721 rooms, Piccadilly; Knightsbridge 2025)	Windowless sleep hotel	Soundproofed, "purified air", no TV, no radio, no window	From £125/night (Time Out; Zedwell)	Marketing only
Nine Hours (Japan)	Capsule chain; 70-capsule Sleep Lab	Bed sensors for movement, snoring, facial images; emailed report; hospital referral	JPY 3,100–9,000/night (Nine Hours; Travel Voice)	Participation only (>100,000 consenting guests)
First Cabin (Japan)	Cabin hotel, 1.2×2.1×1.0 m to 2.1×2.1×2.1 m	Reading light, fan, USB, emergency button; curtain or door	JPY 4,000–10,000 (First Cabin; Unseen Japan)	None
Equinox Hotel New York (Sleep Lab, July 2025)	Luxury sleep rooms with Matthew Walker	Light-absorbent and sound-dampening finishes, measured blackout, smart thermostats, dual-zone temperature mattress, 30-min pre-sleep transition, soundscape wake, AM/PM rituals, cryotherapy	n/f; about €1,500–1,800/night per a secondary guide (Hotel-Online; Business Wire)	Marketing only
MetroNaps / Restworks EnergyPod	Office nap chair	Timed nap, privacy visor, audio, vibration wake	$8,000–13,000 purchase; historically $12–14 per 20-min nap (Unanswered guide; The Points Guy)	None
Chinese capsule OEMs (Guangzhou Wimi, Micane)	Capsule beds ~2×1×1.2 m	Light, fan or HEPA ventilation, power, sometimes TV	$413–800 per unit, MOQ 2 (Accio; SourceReady on Wimi)	None

Three lessons stand out. The premium operators sell exactly the target set from the first section — Zedwell's "dark, quiet, filtered air, no TV, ambient light" and Equinox's "dark, quiet and cool" — rather than gadgets, and Zedwell removed TVs, radios and windows deliberately "so nothing disturbs guests" (Zedwell blog; Criterion Capital), while Japanese budget capsules still ship screens; a cabin network resolves this with a controller tablet that defaults to a media-free sleep mode. Nobody publishes outcome data, so an operator that reports even simple aggregates (median time to settle, share of stays with CO2 held under 800 ppm, measured dB(A)) would be first in the category, with Nine Hours' anonymised research-partnership model as the template. On price, Napcabs (€12–17 per hour, 24/7, two-hour minimum) is the only directly comparable European self-service cabin, US airport rooms sit at roughly €28–45 per hour, capsule nights at €18–60 and Zedwell cocoons around €145; a HAVN two-hour stay at €25–35, overnight at €45–70 and a 20-minute Health Pod session at €25–45 would sit inside those bands.

Priorities per cabin type and questions for Soundbox

Tier	Single Cabin (2.0×2.4 m, bed 200×100)	Double Cabin (2.4×2.4 m, bed 200×160)	Health Pod (2.0×2.4 m, therapy bed, 630–850 nm panel)
Must-have for the Berlin pilot (Nov 2026)	Envelope to DIN 4109 enhanced equivalence (wall ≥52 dB, door ≥37 dB Rw) with duct silencers and a measured ≤30 dB(A) at the pillow; true blackout <1 lux with status LEDs off and hard-off relays for any window or tablet; CO2-controlled ventilation (SCD41 sensor, boost 750 ppm, alarm 1,000 ppm, full-flow reset purge) with HEPA and carbon on supply, no ioniser or ozone; temperature setpoint 18–24 °C (default 20 °C) with cooling path of 300–600 W sensible; four-channel lighting (tunable white 2200–6500 K plus amber/red) on DALI-2 DT8 or Casambi with PstLM ≤1, SVM ≤0.4, measured SPD/mEDI, 30-min dawn ramp to 100–300 lx and floor-level red cue lines; zipped mattress encasement plus 5-minute linen cassette; ENF/E0 panels with REACH Appendix 14 chamber report ≤0.062 mg/m³ and AgBB-2024 report on build-ups; EN 13501-1 B-s1,d0 linings and Cfl-s1 floor from an EU lab; door opens from inside without key or power, external isolator, smoke detector tie-in and written fire-concept sign-off; LVD/EMC/RoHS/RED technical file with CB reports; radar presence sensor for reset and occupancy; sleep-mode controller with no media	All Single Cabin items, plus: ventilation sized for two (≥72 m³/h, 5.7 ACH) with the same setpoints; two independent reading lights; heavier-duvet option per sleeper; a bed count of 2 Gastbetten per cabin reflected in the station plan so the building stays at ≤12 beds under Berlin's BetrVO; no sleep tracking offered for two-person occupancy unless two validated sensors are fitted	Single Cabin environmental and compliance baseline, plus: 20-minute Power Nap and breathing protocols delivered by audio and light with lying time hard-capped at 30 minutes; panel fixed on the long wall at 20–30 cm with a nine-point irradiance map at mounting distance, EN 62471 report including the 780–1400 nm retinal-thermal limit, session timer and per-booking exposure lock, eye-protection acknowledgement and disposable goggles, thermal cut-out and fan-noise spec; ventilation boost ≥100 m³/h during and after sessions; all copy limited to "rest, relaxation, recovery support" with no skin, sleep-disorder, cortisol or longevity claims; manufacturer's declared non-medical intended purpose on file; in-app health screening gate; wipeable surfaces and disposable liners for all contact items; no flashing or strobe light
Should-have for series production	Virtual window: Innerscene Circadian Sky 450×1200 portrait or OEM 46 mm edge-lit tunable light box on the ceiling or high wall, scheduled as daytime light dose and dawn simulator, off during sleep; pink-noise masking speaker, default off; opt-in lavender cartridge diffuser; water-loop bed-cooling pad option; opt-in 60 GHz radar sleep report with explicit consent, no cameras, on-device audio processing, DPIA completed; interlocked 254 nm UV-C reset pass logged per cycle; cove uplight and light matte ceiling for perceived height; published per-stay environment record (CO2, °C, dB)	Dual-zone bed-cooling pad (Eight Sleep-class 13 °C per side as reference spec); larger virtual-window panel budget justified by the higher price point; weighted-blanket option as booked extra with its own wash cycle	HRV/resonance-breathing audio with optional wearable integration; 11-minute yoga nidra and guided mindfulness library; caffeine-before-nap offer at the kiosk; blue-enriched ≥250 mEDI nap-exit light; vibroacoustic transducers in the therapy bed framed as relaxation; opt-in FIR warm mat capped around 40–45 °C surface; OEM 300 W-class panel ($280–600 FOB) once compliance package is proven

Tier	Single Cabin (2.0×2.4 m, bed 200×100)	Double Cabin (2.4×2.4 m, bed 200×160)	Health Pod (2.0×2.4 m, therapy bed, 630–850 nm panel)
Optional / experimental	Rayleigh sun-and-sky skylight (Yeelight P21 at €373 or Innerscene A7) in a roof cut-out as a premium variant where host ceilings exceed 2.5 m; far-UVC 222 nm only with ozone monitoring; in-house pilot study on window-versus-no-window sleep metrics	Head-tracked parallax window (single-viewer limitation makes it unsuitable); weighted blankets for two	PEMF mat (opt-in, pacemaker exclusion enforced by screening); compression boots only in a 60–90 minute booking format; closed-loop acoustic slow-wave stimulation (not recommended: needs EEG and N3 sleep, effect declining); cold exposure, oxygen and grounding excluded

The open questions below are the ones whose answers change the design or the price, in the order HAVN should put them to Mr. Ming.

#	Question for Soundbox	Why it matters
1	Which formaldehyde grade do your boards carry under GB/T 39600-2021 and GB 18580 (E1/E0/ENF), and can you supply a REACH Appendix 14 chamber test report (≤0.062 mg/m³) from an ISO 17025 lab for the finished cabin article?	Legal condition of sale in the EU since 6 August 2026
2	Can you commission a 28-day AgBB-2024 VOC emission test on the assembled wall, ceiling and floor build-ups (not raw board), and a REACH SVHC declaration for foams, adhesives and textiles?	German building law treats AgBB as the health-protection benchmark; needed for any "healthy materials" claim
3	What materials make up your panels today (core, laminate, foam, felt, floor), and which fire reports do you hold — GB 8624 or EN 13501-1 from an EU-accredited lab? Can you reach B-s1,d0 linings and Cfl-s1 floor?	Berlin fire-concept sign-off and insurer acceptance
4	What Rw (EN ISO 10140-2) do your wall and door achieve, and what ISO 23351-1 class? Can the door reach ≥37 dB with both ventilation penetrations silenced, and what is the sound power of the Panasonic unit at 36 and 180 m³/h?	The 30 dB(A) target is door- and duct-limited; station sites need ~10 dB more than office pods
5	Which Panasonic ERV model is specified: heat-recovery efficiency, CO2-demand input, filter options (HEPA + carbon), and power at low speed?	CO2 setpoints, noise floor and thermal load all depend on it
6	Can an 84 mm-deep luminaire (Circadian Sky 450×1200 portrait) or a 46 mm edge-lit box be integrated within one 80 cm panel module without compromising acoustic performance, and can the roof panel take a 200–270 mm Rayleigh unit cut-out?	Determines which virtual-window option is feasible and where
7	Will you build the four-channel lighting on DALI-2 DT8 or Casambi (Sunricher/LEDGEAR), and can you deliver measured SPD and mEDI per channel plus driver flicker data (PstLM, SVM)?	Circadian recipe and Ecodesign-equivalent compliance
8	For the red-light panel: supplier, chip count, drive current, nine-point irradiance map at 20 cm, EN 62471 report including IR retinal-thermal class, fan noise, surface temperature, timer/interlock design, and the panel maker's declared intended purpose (non-medical)?	MDR Annex XVI exposure, eye safety and heat in a 10.6 m³ room
9	Who will be manufacturer of record for CE? Will Soundbox supply CB-scheme reports (IEC 60335/62368/60598), a whole-cabin EMC test, a component bill with declarations, RoHS and RED documentation — or does HAVN brand and take the technical file?	Decides liability, record-keeping and the authorised-representative question

#	Question for Soundbox	Why it matters
10	What cooling do you integrate (split, water loop, Peltier) for 300–600 W sensible load, and what is the heat-rejection path into the host building?	Thermal target of 18–24 °C cannot be met by ventilation alone in warm host spaces
11	Ceiling provision for sprinkler drop heads or pass-through and a smoke-detector mount; electronic door lock with fail-safe release on power loss; external main isolator?	Non-negotiable for Berlin fire-concept approval
12	Reset design: mattress encasement and linen-cassette fixing, interlocked 254 nm UV-C module, radar occupancy sensor, controller platform (Tuya Zigbee, ESP32 or Matter) and API for HAVN's booking system?	The 20-minute reset is a logistics-and-interlock design
13	Lead time, MOQ and price breaks at 11, 100 and 500 cabins per type, and confirmation that the HAVN mark is registered in China before tooling starts?	Programme planning and brand protection

Conclusion

The research reframes HAVN's product from "a cabin with health technology inside" to "a cabin that measurably delivers the four conditions the body needs to sleep, with light as its one active ingredient". That shift has commercial consequences: it moves money from branded gadgets toward the door, the ducts, the drivers and the test reports, and it makes the per-stay environment record — CO2 held under 800 ppm, 30 dB(A) at the pillow, lux at wake — the differentiator no operator in the category yet publishes. The Health Pod survives this scrutiny, but as a nap-and-breathing room with a red-light module framed as recovery support; the "longevity" story rests on sleep regularity and short naps, not on any device, and HAVN should resist the temptation to let the panel's marketing outrun a 20-athlete cohort study.

For the Soundbox relationship the implication is that compliance documentation is now a product feature rather than paperwork. The formaldehyde limit is already law, the fire classes decide whether a Berlin building accepts the cabins, the CE file decides who is liable, and the intended-purpose wording decides whether the Health Pod is a wellness room or an unregistered class IIb medical device. A manufacturer who can answer the thirteen questions above with test reports rather than labels is worth more to HAVN's 2,500-cabin plan than one who can build a cheaper box, and the follow-up email to Mr. Ming should say exactly that.